Milestone 01

PDF edition of preserved repository document | docs/milestone_01.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

First milestone report

Implemented

Phase 0: Git main branch, installable Python 3.11+ package, MIT license, TOML configuration loader, documentation, Makefile, optional lint/type/test tools and a CI test workflow.

Phase 1: frozen Entity, State, Relation, Action, Event, Result, Observation, Goal, Transition and Trajectory records. Five distinct evidence categories.

Phase 2: deterministic decimal digit/carry multiplication, partial-row accumulation, signed inputs and zero, required environment API and trajectory recording.

Phase 3: full transition replay verification, independent final-product oracle and goal evaluation. Illegal actions are rejected without modifying the state.

Tested

Python 3.12.14 standard-library unittest runner: 21 tests passed. Exhaustive oracle checks cover all 1,681 pairs in $[-20,20]^2$; 200 seeded pairs cover larger signed operands. Also tested carries, shifted partial rows, input/action validation, immutability, reset, configuration, and tampered/reordered/duplicated/incomplete trajectories. Seeded oracle checks are tests, not generated benchmark datasets. pytest, Ruff and MyPy were not installed or run. CI configuration exists but its hosted execution has not been observed. No neural experiments were performed.

Verified demonstration: 19 × 3

Initial digits (least significant first): [9,1] and [3]. Position 0, carry 0, row value 0, accumulated value 0. Goal: produce the correct signed integer product.

Action	Event	Actual result	Position after	Carry after	Row after	Total after	Phase after
multiply	digit_written	27	1	2	7	0	multiply
multiply	digit_written	5	2	0	57	0	flush
flush	carry_flushed	57	2	0	57	0	accumulate
accumulate	row_accumulated	57	0	0	0	57	finish
finish	answer_materialized	57	0	0	0	57	done

First digit: 9×3+0=27 → write 7, carry 2. Second: 1×3+2=5 → row 57, carry 0. Flush carry, accumulate row and materialize answer. Final state: done, answer 57, accumulated 57, carry 0. Trajectory valid=true, final answer valid=true, goal reached=true. Every before/after state, action, event and actual result is saved in examples/19_times_3.json.

Not yet implemented

Dataset generation/splits and their reproducibility/OOD tests; benchmark metric calculations; model/backbone interfaces and baselines; learning regimes; memory; data-efficiency studies; operational ablations; historical reference collection; simulation; second domain; UI and multimodal components. Placeholders are scaffolding.

Known limitations

The policy is a prescribed algorithm with one valid next action; no learned procedure or cognitive superiority is demonstrated. Replay shares environment transition logic; the independent product oracle and broad arithmetic tests mitigate shared bugs. Input observation is authoritative and the verifier cannot authenticate an external user's original question. Only in-memory typed records are replayed; JSON import and schema migration are not implemented. Dataclass annotations are not universal runtime validation, though input boundaries and legal actions are validated. Large operands are subject to Python integer-string conversion and resource limits. Partial-row addition is an exact integer operation rather than an addition-domain

trajectory. Local runtime was 3.12; the configured 3.11 CI has not yet run.

Relevant repository tree

```
mindscape/
■■■ README.md, LICENSE, pyproject.toml, Makefile
■■■ configs/{base,multiplication}.toml
■■■ docs/{architecture,research_hypotheses,benchmark,experimental_protocol}.md
■■■ docs/decisions/001-foundation.md
■■■ examples/19_times_3.json
■■■ src/mindscape/core/{schema,config,entities,states,...}.py
■■■ src/mindscape/environments/base.py
■■■ src/mindscape/environments/multiplication/environment.py
■■■ src/mindscape/verification/verifier.py
■■■ scripts/run_episode.py
■■■ tests/test_foundation.py
■■■ .github/workflows/tests.yml
```

Git

Branch main. Logical commits:

- d3d2261 repository and research specification
- a991db1 typed schemas and configuration
- 963dec9 environment and verifier
- ac51554 tests and recorded demonstration

This report is committed in a final documentation commit. Final status is checked separately after committing.